What is DRAM?

Dynamic Random Access Memory / Volatile semiconductor memory

01 Basic structure

DRAM temporarily stores, at high speed, the data processed by CPUs and GPUs. It holds information as electrical charge and requires periodic refresh operations. It is volatile memory: stored data disappear when power is switched off.

02 DRAM versus NAND flash

Feature	DRAM	NAND
Main use	Working memory	SSDs and storage
Power off	Data lost	Data retained
Access speed	Fast	Slower than DRAM
Cost per unit of capacity	Generally higher	Generally lower

03 Representative DRAM standards

DDR5: A main-memory standard for PCs and servers.
LPDDR5X: A low-power standard for smartphones and thin devices.
HBM: Stacked DRAM that provides high data-transfer bandwidth.

The DRAM industry

AI demand, major manufacturers and market cycles

04 AI and HBM

Generative AI training and inference require large volumes of data to reach GPUs at high speed. HBM is placed close to the GPU to provide wide memory bandwidth. If data delivery cannot keep up, the GPU’s computing power cannot be fully used.
AI processing: GPU computation supported by high-speed HBM data delivery.

05 Major DRAM manufacturers

Company	Location	Activities
Samsung Electronics	South Korea	Broad DRAM and NAND business
SK hynix	South Korea	High-performance memory, including HBM
Micron Technology	United States	DRAM, HBM and NAND
Nanya Technology	Taiwan	Primarily conventional DRAM

06 Industry characteristics and price cycles

Oligopoly: Three major companies supply most of the world’s DRAM.
Supply and demand: Oversupply pushes prices down; tight supply pushes prices up.
Capital expenditure: Shrinking features and expanding capacity require substantial investment.
Production allocation: Expanding HBM output can also affect conventional DRAM supply.